

RESEARCH ARTICLE

Mixture analysis using non-negative elastic net for Raman spectroscopy

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Abstract

Raman spectrum is wealth of structural information, which can be used as molecular fingerprint to identify compounds. There are relations between the intensities of Raman peaks and the concentrations of compound, and they can be used to estimate relative concentrations of the compounds in mixture. However, it is still challenging to interpret the Raman spectrum of mixture. The non-negative lasso (NN-LASSO) has been used to solve this problem, but it failed to identify highly correlated compounds. The quadratic term of non-negative elastic net (NN-EN) can ensure the stability of the fitted model. Therefore, a novel mixture analysis method was developed based on NN-EN in this study. It has been applied to analyze the simulated, liquid, powder, tablet, and quantitative mixture datasets. Results showed that NN-EN can identify the compounds in mixture with high accuracy and estimate their relative concentrations with small deviation. Furthermore, NN-EN was more stable than NN-LASSO when the spectra of some compounds are highly correlated. It is a promising approach for analyzing Raman spectra of mixtures.

KEYWORDS

mixture analysis, non-negative elastic net, Raman spectroscopy, spectral database

1 | INTRODUCTION

Raman spectrometer collects the scattered light from sample illuminated by laser. The scattered light is shifted in frequency because of the interaction between the light and the molecular vibrations. Therefore, Raman spectroscopy is commonly used in chemistry to retrieve vibrational information of molecules and used as molecular fingerprint to identify the molecules. It has the advantages of high speed, nondestructive, and sample preparation free with widely applications in artwork,¹ drug screening,^{2,3} forensics,^{4–6} online industrial process monitoring,⁷ food analysis,⁸ law enforcement,⁹ and explosives detection.^{10–15} Raman applications heavily rely on the statistical comparison of the spectra of unknown samples with the spectra in library. Therefore, numerous chemometric methods have been proposed.

Chemometric methods for Raman spectroscopy include preprocessing,^{16–23} pattern recognition,^{24–27} multivariate calibration,^{28,29} and instrument standardization.^{30–35} There are fluorescent backgrounds and noises in Raman spectra of complex samples, so preprocessing is needed before further statistical analysis. Spectral preprocessing can eliminate the fluorescent background and noise. Modern Raman spectrometers can generate large amounts of spectra within a short time. Pattern recognition and multivariate calibration are often applied to extract useful chemical information

from these spectra. In order to use the pattern recognition and calibration models extensively, instrument standardization is crucial to avoid the costly and time-consuming recalibration.

Library search is also essential to identify compounds using Raman spectra. Most of the library search methods are mainly used for the pure compounds. However, the samples are mixtures in practical applications. Researchers are in urgent need of mixture analysis methods for Raman spectroscopy. Zavaleta et al.³⁶ and Keren et al.³⁷ proposed the classical least squares method, which could estimate the relative concentrations under the hypothesis that there are linear additive relationships between spectrum of mixture and spectra of pure components. Zhang and coworkers³⁸ introduced the non-negativity constraint for least squares when analyzing Raman spectrum of mixture. In addition, Li et al. focused on the regression methods in the quantitative analysis of Raman spectroscopy.³⁹ The LASSO algorithm can perform shrinkage and variable selection and build the calibration model simultaneously.⁴⁰ Recently, NN-LASSO has been applied in analyzing Raman spectrum of mixture by imposing the non-negativity constraint to regression coefficient.⁴¹

In this study, a novel method has been proposed to solve the mixture analysis problem based on non-negative elastic net (NN-EN). It can simultaneously identify compounds in mixture and estimate their relative concentrations. In order to verify the feasibility of NN-EN for mixture analysis, simulated spectra with high fluorescence backgrounds were generated and analyzed by NN-EN. Then, liquid and powder mixtures with different relative concentrations were used to evaluate the compound identification capabilities of the NN-EN method. The tablet dataset was used to test the performance of NN-EN in practical application. Furthermore, 94 mixtures with different concentrations in the quantitative dataset were analyzed to benchmark the relative concentration estimation capabilities of the NN-EN method. Finally, the advantages of the NN-EN method were demonstrated by comparing NN-EN with the NN-LASSO method.

2 | THEORY

The theory of NN-EN was briefly introduced in this section. The noises and backgrounds were removed from Raman spectrum by Whittaker smoother and adaptive iteratively reweighted penalized least squares (airPLS) method. Then, the NN-EN was used to identify compounds in the mixture and estimate their relative concentrations. Standardization methods for spectra were also applied to avoid the costly and time-consuming remeasurement of spectral database. Figure 1 is an overview of the proposed method. The techniques used in NN-EN are explained as clearly as possible in the following sections.

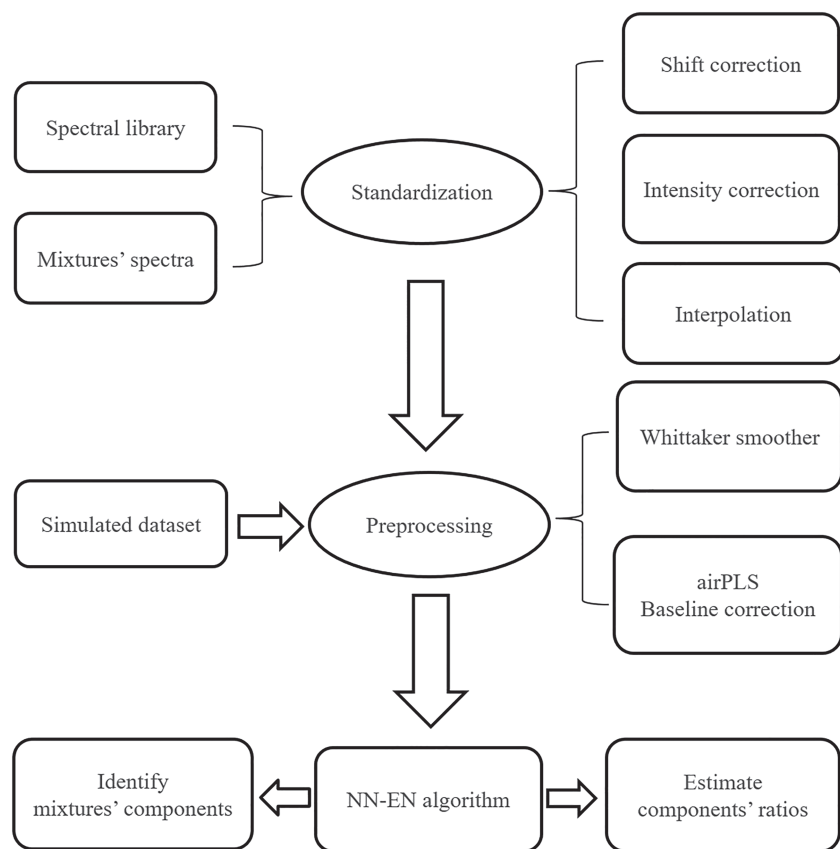


FIGURE 1 Overview of the proposed mixture analysis method for Raman spectra

2.1 | Spectral preprocessing methods

Raman spectrum can be regarded as the combination of analytical signals, background, and noises. Preprocessing methods were needed to correct background and eliminate noises. The Whittaker smoother was used to eliminate noises because it can control smoothness continuously.⁴² The smoothed Raman spectrum can be obtained by solving the following linear equation:

$$\mathbf{z} = (\mathbf{I} + \lambda \mathbf{D}^T \mathbf{D})^{-1} \mathbf{x}, \quad (1)$$

where $\mathbf{x}$ is the raw Raman spectrum and $\mathbf{z}$ is the smoothed Raman spectrum, and they are column vectors with n elements. $\mathbf{I}$ is the identity matrix. $\mathbf{D}$ is the differential matrix which can calculate the numeric difference $\Delta \mathbf{z}$ by multiplying $\mathbf{z}$ with $\mathbf{D}$ ($\Delta \mathbf{z} = \mathbf{D} \mathbf{z}$), and the row and column of matrix $\mathbf{D}$ are $n - 1$ and n respectively. λ is the smoothness parameter, and it can be optimized by users.

The background of Raman spectrum was corrected using airPLS.⁴³ It is an iterative method, and each step can be presented by the following form:

$${}^tQ = \sum_{i=1}^n {}^t w_i |x_i - {}^t z_i|^2 + \lambda \sum_{j=2}^n |{}^t z_j - {}^t z_{j-1}|^2. \quad (2)$$

Here, Q is the objective function. x_i and z_i are the i th elements of $\mathbf{x}$ and $\mathbf{z}$ respectively. n is the number of elements in the spectral vector $\mathbf{x}$ and $\mathbf{z}$. t is the iteration number. λ is the smoothness parameter. The weight vector $\mathbf{w}$ can be calculated adaptively using an iterative procedure. The weight vector $\mathbf{w}$ was initialized as ${}^0 \mathbf{w} = 1$. After initialization, $\mathbf{w}$ of each iterative step, t could be obtained using following equation:

$${}^t w_i = \begin{cases} 0 & x_i \geq {}^{t-1} z_i, \\ e^{\frac{{}^{t-1} z_i - x_i}{|{}^t d|}} & x_i < {}^{t-1} z_i. \end{cases} \quad (3)$$

Vector ${}^t \mathbf{d}$ contains negative elements of difference between $\mathbf{x}$ and ${}^{t-1} \mathbf{z}$ in t iteration step. The λ parameter can be adjusted by users.

In this study, acceptable smoothing and baseline correction results were obtained with *differential order* = 1 and $\lambda = 10$, and they were optimized using the spectra in the library and the C++ version airPLS with graphical user interface (Figure S1). The Raman spectrum of 1,4-benzoquinone was chosen to demonstrate the necessity of preprocessing. The Raman spectra before and after preprocessing were shown in Figure 2. The Whittaker smoother method can reduce the noises effectively, and the airPLS algorithm can correct the background flexibly.

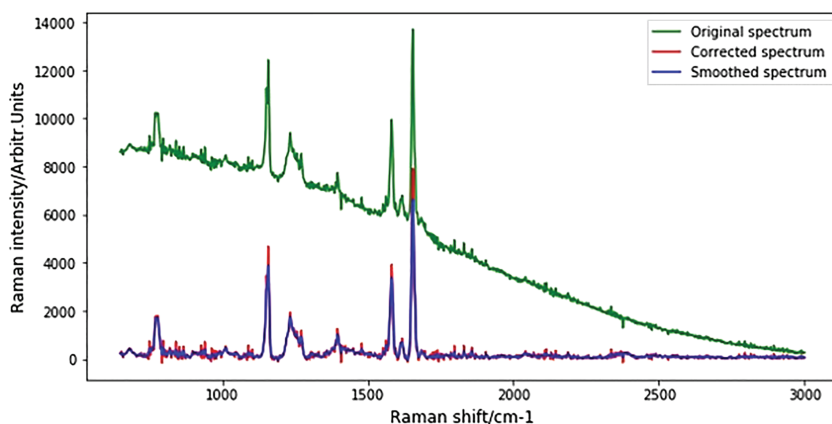


FIGURE 2 Baseline correction and smoothing results of the Raman spectrum of 1,4-benzoquinone by airPLS and Whittaker smoother, and the green line represents the original spectrum, the red line represents the corrected spectrum, and the blue line represents the smoothed spectrum

2.2 | NN-EN for mixture analysis

In this section, the NN-EN algorithm was elucidated for compound identification and their relative concentration estimation. Because LASSO algorithm tends to select only one variable from a group of correlated variables, Zou and Hastie proposed the elastic net (EN) algorithm in 2005.⁴⁴ It can solve the problem of regression instability when the number of variables increases. The EN algorithm can be expressed with the following equation:

$$\arg \min_{\mathbf{r}} \|\mathbf{y} - \mathbf{X}\mathbf{r}\|_2 + \lambda_1 \|\mathbf{r}\|_1 + \lambda_2 \|\mathbf{r}\|_2. \quad (4)$$

Here, $\mathbf{y} \in R^n$ is the Raman spectrum of a mixture. $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m)$ with $\mathbf{x}_i \in R^n$ represented the Raman spectrum of the i th component in the mixture, and $\mathbf{r}_i$ represents its relative concentration. λ_1 and λ_2 were the regularization parameters, and they should be larger than zero. The EN algorithm can find the relative concentrations to minimize the fitting error. Besides, the concentration of the compound should be non-negative, so non-negative restrictions were required. In 2016, Mandal and Ma⁴⁵ proposed the multiplicative iterative path method. The fitting error can be expressed as

$$\phi(\mathbf{r}) = \|\mathbf{y} - \mathbf{X}\mathbf{r}\|_2 + \lambda_1 \|\mathbf{r}\|_1 + \lambda_2 \|\mathbf{r}\|_2, \quad (5)$$

using Karush–Kuhn–Tucker conditions to make $\mathbf{r}$ non-negative:

$$\begin{aligned} \frac{\partial \phi}{\partial \mathbf{r}_i} &= 0 \quad \text{if } \mathbf{r}_i > 0, \\ \frac{\partial \phi}{\partial \mathbf{r}_i} &< 0 \quad \text{if } \mathbf{r}_i = 0. \end{aligned} \quad (6)$$

The NN-EN algorithm can be written as

$$\arg \min_{\mathbf{r}} \|\mathbf{y} - \mathbf{X}\mathbf{r}\|_2 + \lambda \{ \alpha \|\mathbf{r}\|_1 + (1 - \alpha) \|\mathbf{r}\|_2 \}, \hat{\mathbf{r}} \geq 0. \quad (7)$$

Here, $0 \leq \alpha \leq 1$. In this study, EN parameter α was set as 0.96 according to Figure S2, which was optimized using the grid search method and the simulated mixture spectra. The calculation times for parameters optimization were listed in Table S1. The pseudocode of NN-EN is as follows:

Step 1: Specify $\lambda_{\max}$, $\lambda_{\min} = \max(\mathbf{X}^T \mathbf{y})$, 10^{-4} , number of steps $n = 100$, set $\lambda = \lambda_{\max}$, $\mathbf{r}^{(0)} > 0$.

Step 2: Compute step size $s = (\lambda_{\max} - \lambda_{\min})/n$.

Step 3: Repeat steps while $\lambda \geq \lambda_{\min}$.

Repeat steps for $k = 0, 1, 2, \dots$, till convergence.

1. Evaluate $\phi(\mathbf{r}^{(k)})$, set $\omega^{(k)} = 1 (0 < \omega^{(k)} \leq 1)$.
2. Compute $\mathbf{r}^{(k)} + \omega^{(k)} s_j^{(k)} \partial \phi(\mathbf{r}^{(k)}) / \partial \mathbf{r}_j$, evaluate $\phi(\mathbf{r}^{(k)} + \omega^{(k)} s_j^{(k)} \partial \phi(\mathbf{r}^{(k)}) / \partial \mathbf{r}_j)$.
3. Repeat (2) until

$$\begin{aligned} \phi(\mathbf{r}^{(k)} + \omega^{(k)} s_j^{(k)} \partial \phi(\mathbf{r}^{(k)}) / \partial \mathbf{r}_j) &\geq \\ \phi(\mathbf{r}^{(k)}) + \xi \omega^{(k)} \left(\partial \phi(\mathbf{r}^{(k)}) / \partial \mathbf{r} \right)^T s_j^{(k)} \partial \phi(\mathbf{r}^{(k)}) / \partial \mathbf{r}_j. \end{aligned}$$

Here, $\omega^{(k)} = \sigma \omega^{(k)}$ for each repetition, σ and ζ are in (0,1) with preset value.

The estimated coefficients be $\mathbf{r}^*$.

Set $\lambda = \lambda - s$, $\mathbf{r}^{(0)} = \mathbf{r}^*$.

2.3 | Component was not present in the library

Considering that a component of the mixture was not in the library. The unknown component was marked as $\mathbf{o} \in R^n$ because the component was outlier relative to the mixture. Assume the component was not in the library rarely, that is, $\mathbf{o}$ was sparse. In this case, $\mathbf{o}$ and $\mathbf{r}$ can be solved by changing the formula 7 as

$$\arg \min_{\mathbf{r}, \mathbf{o}} \|\mathbf{y} - \mathbf{X}\mathbf{r} - \mathbf{o}\|_2 + \lambda \{ \alpha \|\mathbf{r}\|_1 + (1 - \alpha) \|\mathbf{r}\|_2 \} + \beta \|\mathbf{o}\|_1, \hat{\mathbf{r}} \geq 0. \quad (8)$$

The β was a parameter for $\mathbf{o}$. Through this formula, $\mathbf{r}$ and $\mathbf{o}$ can be solved. It may solve the problem that the components of the mixture were not in the library.

2.4 | Standardization for spectra

There might exist systematic difference between Raman spectrometers.⁴⁶ For this reason, the Raman spectrum should be standardized toward the spectra in the library before statistical comparison. Spectral standardization contains shift alignment and intensity correction.^{30,35}

In order to align the Raman shift displacements between spectrometers, Raman spectra of standard compounds should be acquired on the primary and secondary spectrometers. The major peaks of compounds can be detected by peak detection methods based on continuous wavelet transform (CWT).^{47–49} Raman shift displacements of these peaks between spectrometers can be fitted using a polynomial. Once the relationship between the primary and secondary spectrometers has been established, the spectrum acquired by the secondary can be standardized toward the primary spectrometer (spectral library) easily. The schematic of Raman shift correction was shown in Figure 3.

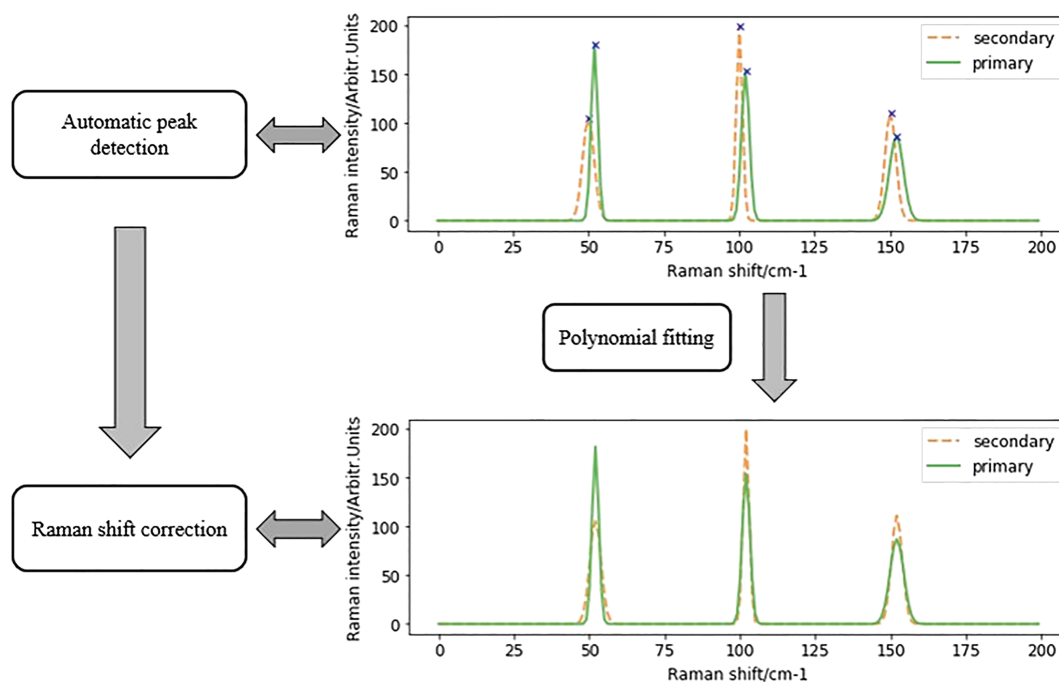


FIGURE 3 Schematic diagram of Raman shift correction for instruments standardization using simulated data. The automatic peak detection method is performed for the primary and the secondary spectra (the yellow line indicates the secondary spectrum, and the green line indicates the primary spectrum), and the transfer curve can be established by polynomial fitting. Finally, the Raman shift is corrected

For intensity correction, standard reference materials 2241 (SRM 2241) was the NIST certified spectroscopic standard for spectrometers employing 785-nm excitation laser.⁵⁰ The SRM 2241 was illuminated by laser and generate a broadband spectrum, which can be described as a fifth-order polynomial. For each spectrometer, the polynomial and the spectrum of SRM 2241 were combined to perform the intensity correction.

3 | EXPERIMENTAL

3.1 | Simulated library and mixture dataset

To verify feasibility of NN-EN method, simulated spectra were generated as the spectral library ($\mathbf{X}_{n \times m}$ where spectral channels $n = 1000$, number of spectra $m = 500$). Each simulated spectrum contained four or five Lorentzian peaks, background, and random noises, and some of the spectral peaks partially overlap. Three simulated spectra were chosen from the library to generate simulated mixture spectra at three different relative concentrations. Background subtraction and noise elimination were performed before the NN-EN analysis.

3.2 | Raman spectral library

Standard compounds were purchased from Changsha Fufan Trade Ltd, which were used to generate the spectral library ($\mathbf{X}_{n \times m}$ where spectral channels $n = 881$, number of spectra $m = 167$). Raman spectra of these compounds were acquired using B&W Tek i-Raman with BWSpec 3.27. The integral time of each Raman spectrum in library was optimized to achieve the best spectral quality. Therefore, their integral times were different, which may lead to the inaccuracy in relative concentration estimation. In order to solve this problem, the intensity of each spectrum was adjusted toward the spectrum of mixture using following equation:

$$\mathbf{x}_u = \mathbf{x} \frac{t_m}{t}. \quad (9)$$

Here, $\mathbf{x}$ is the spectrum in library. $\mathbf{x}_u$ is the adjusted spectrum. t_m and t are the integral times of mixture spectrum to analyze and spectrum in the library, respectively.

3.3 | Liquid and powder mixture datasets

Three relative concentrations of methanol, ethanol, and acetonitrile of liquid mixtures (4:3:3, 5:3:2, and 7:2:1) and polyacrylamide, sodium acetate, and sodium carbonate of powder mixtures (polyacrylamide:sodium acetate = 1:1, polyacrylamide:sodium acetate:sodium carbonate = 5:3:2, and polyacrylamide:sodium acetate:sodium carbonate = 1:1:1) were measured by the B&W Tek i-Raman spectrometer and used to test whether NN-EN can identify the components in these mixtures correctly.

3.4 | Tablet dataset

Spectra of three tablets including Excedrin Migraine, Consumer Value Stores (CVS) Migraine, and Alka Seltzer Plus Night have been acquired and used to test the practicality and effectiveness of NN-EN. Excedrin Migraine tablet contained three components of acetaminophen, hydrocortisone, and niacinamide. CVS Migraine tablet contained three components of acetaminophen, niacinamide, and dopamine HCl. Alka Seltzer Plus Night tablet contained three components of L-lysine acetate, 2-propanol, and niacinamide. The pure material spectra of these components were in the spectral library.

3.5 | Quantitative mixture dataset

Ternary mixtures of different volumes of methanol, acetonitrile, and distilled water were used for evaluating the relative concentration estimation performance of NN-EN. We acquired Raman spectra of 94 mixtures with B&W Tek i-Raman spectrometer and BWSpec 3.27. All the spectral integration times were 7.5 s.

3.6 | Computing facilities and implementation

In this study, spectral file parse, integration time unification, linear interpolation, spectral normalization, and spectral standardization were implemented using Python 3.5.4. Spectral preprocessing method such as smoothing, background correction, were performed using airPLS 3.0.0 (https://github.com/zmzhang/airPLS_R) and nnlasso 0.3 (<http://CRAN.R-project.org/package=nnlasso>) packages in R 3.4.3. The computer operating system was Windows 10 with an Intel (R) Core (TM) i7-4710MQ processor and 32 GB memory.

4 | RESULTS AND DISCUSSION

4.1 | Results of simulated dataset

The results of simulated mixtures by NN-EN were shown in Figure 4. Each row represented a simulated mixture spectrum and results of NN-EN. The first, second, and third columns represented pure spectrum, mixture spectrum, and regularization path of NN-EN, respectively. The black line represented the original spectrum, and the red line represented the preprocessed spectrum. The Whittaker smoother and airPLS can eliminate the noises and correct the background effectively. In the regularization path, only the coefficients of compounds in the simulated mixture

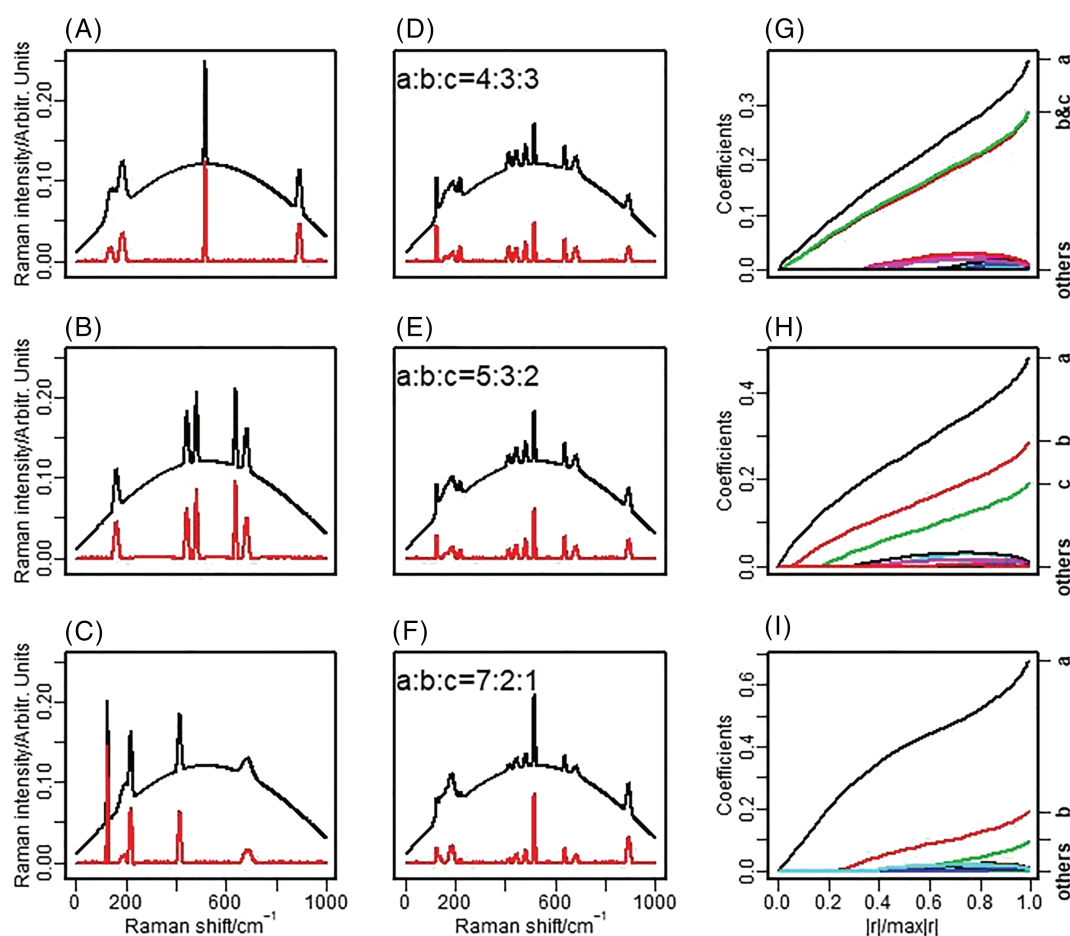


FIGURE 4 Results of non-negative elastic net (NN-EN) on the simulated dataset. The original spectra (black line) and the corrected spectra (red line) after preprocessing of three different components are shown in a, b, and c, respectively. The second column represents the original spectra (black line) and the pretreated spectra (red line) by mixing the three pure components at 4:3:3, 5:3:2, and 7:2:1, respectively. The third column represents the regularization paths of NN-EN. The black, red, and green lines represent the regularization paths of three pure components, respectively. The value at the end of regularization path represents the final predicted relative concentration

were significantly larger than zero. It means that NN-EN can identify the components in mixture from library. One can also see that the values of coefficients of NN-EN are the same as their relative concentrations. It means that NN-EN can estimate the relative concentration of each identified compound. NN-EN is feasible to identify the components in mixture and estimate their relative concentrations for the simulated Raman spectrum with complex background and noises.

4.2 | Results of liquid and powder dataset

The results of NN-EN on liquid and powder dataset were shown in Table 1. Six spectra of mixtures were analyzed by the NN-EN, and the components in the mixture were successfully identified. The true positive (*TP*), false positive (*FP*), and true positives rate ($TPR = TP/(TP+FN)$) have been used to evaluate the performance of identification. *TP* represents the number of mixture components that have been correctly identified. *FP* represents the number of incorrectly identified components. *FN* represents the compound in the mixture but fail to identify by NN-EN or NN-LASSO. The NN-LASSO algorithm produced the *FN* results, and the NN-EN algorithm can identify all the compounds in these six mixtures correctly. The reason is that NN-LASSO tends to select only one compound from a group of similar compounds. On the contrary, the NN-EN algorithm tends to select all the similar compounds in the group. Though some *FP*s may be introduced by NN-EN, they can be eliminated easily based on their relative concentrations. The regularization paths of the NN-EN and NN-LASSO for liquid mixtures were shown in Figure 5. The first and second rows represented the results of NN-EN and NN-LASSO. a, b, and c represented methanol, ethanol, and acetonitrile, respectively. It can be seen that the liquid mixture components in the relative concentration of 4:3:3 and 5:3:2 can be accurately estimated by both NN-EN and NN-LASSO. Although NN-EN produced *FP*s on the mixture of 4:3:3, the relative concentration of *FP* was very low and can be eliminated easily by setting a relative concentration threshold. For the 7:2:1 liquid mixture, NN-LASSO cannot identify the component with lowest relative concentration, acetonitrile, and NN-EN can identify it correctly due to the grouping effect of EN regularization.

4.3 | Results of tablet dataset

The identification results of the three tablets were listed in Table 2. As can be seen from this table, the active pharmaceutical ingredients (APIs) of each tablet were identified correctly. In addition, Figure 6A–C showed the identification results of Excedrin Migraine, CVS Migraine, Alka Seltzer Plus Night, respectively. Each subgraph corresponded to a, b, and c, respectively, indicating the APIs of each tablet. In Figure 6A, a is acetaminophen, b is hydrocortisone, and c is niacinamide. In Figure 6B, a is dopamine HCl, b is niacinamide, and c is acetaminophen. In Figure 6C, a is niacinamide, b is 2-propanol, and c is L-lysine acetate. It can prove the practicability and effectiveness of the NN-EN method.

TABLE 1 Identification results of different relative concentrations of ternary mixture

Mixtures		True positives rate (TPR)		False positive (FP)	
		NN-EN	NN-LASSO	NN-EN	NN-LASSO
Liquid	1 (433)	3/3	3/3	1	1
	2 (532)	3/3	3/3	0	0
	3 (721)	3/3	2/3	0	0
Powder	4 (11)	2/2	1/2	2	0
	5 (532)	3/3	2/3	0	0
	6 (111)	3/3	1/3	0	0

Abbreviations: NN-EN, non-negative elastic net; NN-LASSO, non-negative lasso.

Note: The numbers in parentheses indicate the relative concentrations of the mixture, respectively.

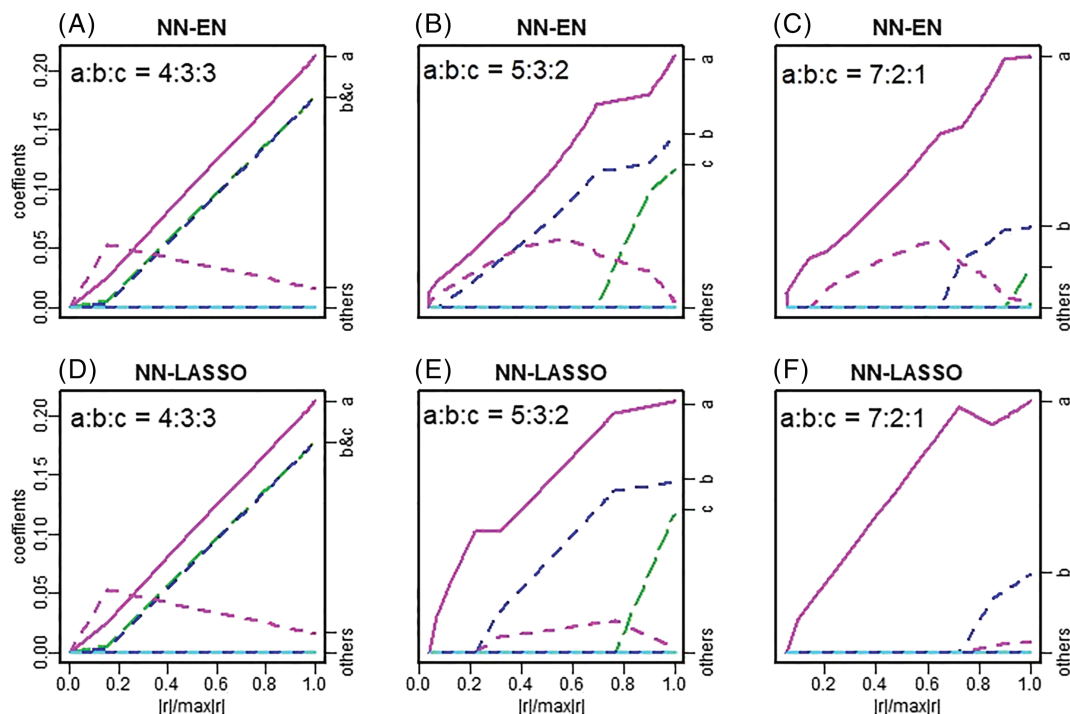


FIGURE 5 Comparison of non-negative elastic net (NN-EN) and non-negative lasso (NN-LASSO) in liquid dataset. Each subplot represents the results of a mixture of methanol (a), ethanol (b), and acetonitrile (c) at 4:3:3, 5:3:2, or 7:2:1. The red, blue, and green lines represent the regularization paths of the three components of a, b, and c, respectively. The value at the end of regularization path represents the final predicted relative concentration

TABLE 2 Results of practical tablet dataset by non-negative elastic net (NN-EN)

Number	Mixture name	Component name	Identification result
(A)	Excedrin Migraine	a (Acetaminophen)	✓
		b (Hydrocortisone)	✓
		c (Niacinamide)	✓
(B)	CVS Migraine	a (Dopamine HCl)	✓
		b (Niacinamide)	✓
		c (Acetaminophen)	✓
(C)	Alka Seltzer Plus Night	a (Niacinamide)	✓
		b (2-Propanol)	✓
		c (L-Lysine acetate)	✓

4.4 | Results of quantitative mixture dataset

The consistency between the predicted relative concentrations and the true relative concentrations can be evaluated by coefficient of determination (also called R^2):

$$R^2 = 1 - \frac{\sum (r - \hat{r})^2}{\sum (r - \bar{r})^2} \quad (10)$$

The percent relative error (PRE) of all samples and root mean square error of prediction (RMSEP) were also used to evaluate the relative concentration estimation performance:

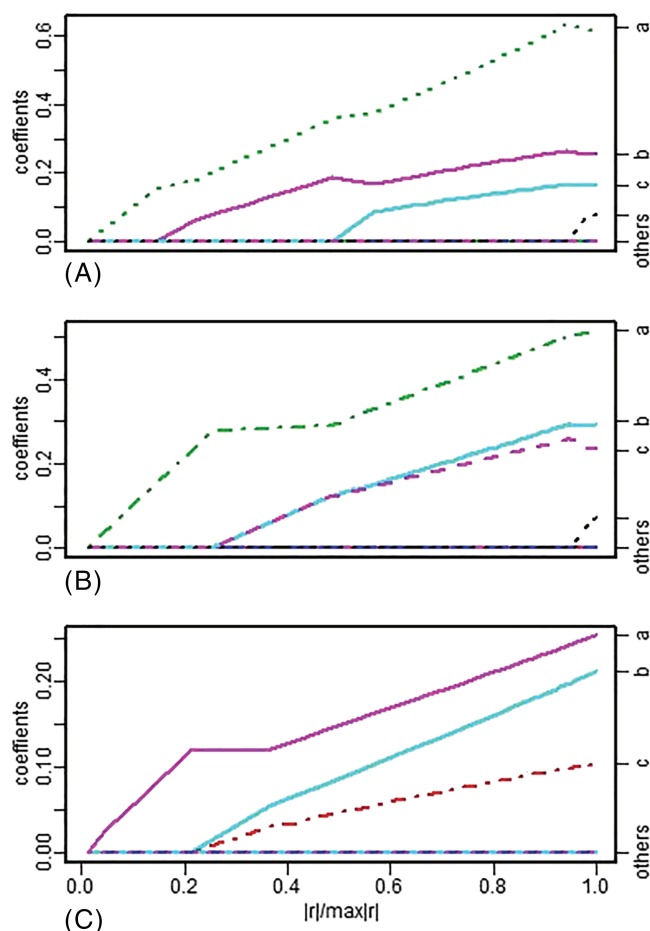


FIGURE 6 (A–C) The recognition results of Excedrin Migraine, CVS Migraine, and Alka Seltzer Plus Night, respectively. Each subgraph corresponds to a, b, and c, respectively, indicating the composition of each tablet. In (A), a is acetaminophen, b is hydrocortisone, and c is niacinamide. In (B), a is dopamine HCl, b is niacinamide, and c is acetaminophen. In (C), a is niacinamide, b is 2-propanol, and c is L-lysine acetate

$$PRE = (1 - R^2) \times 100\%, \quad (11)$$

$$RMSEP = \sqrt{\frac{\sum (\hat{r} - r)^2}{n}}, \quad (12)$$

In Equations 10–12, r and $\hat{r}$ represented the true and predicted relative concentration, $\bar{r}$ represented the mean of the true relative concentration. n represented the number of spectra.

Results of 94 ternary mixtures in quantitative mixture dataset were listed in Table S2, including the methanol relative concentrations, the acetonitrile relative concentrations, the predicted relative concentrations of methanol, and the PREs of NN-EN and NN-LASSO. The predicted and the true relative concentrations of methanol in noninterferences (without acetonitrile), low interferences (small acetonitrile relative concentrations), and high interferences (large acetonitrile ratios) were shown in Figure 7A–C, respectively. The predicted relative concentrations were consistent with the true relative concentrations because of high R^2 . The PRE and RMSEP were also small for NN-EN method. In addition, the PLS and multivariate curve resolution alternating least square (MCR-ALS) methods were used for comparison with NN-EN on the quantitative mixture dataset, and the results of the analysis were shown in Table 3. The results showed that NN-EN is comparable with PLS and MCR-ALS. It means that NN-EN can estimate the relative concentrations accurately and is not sensitive to interferences.

4.5 | Advantages of non-negative EN for mixtures analysis

NN-EN algorithm was used for analyzing Raman spectra of mixtures. It can shrink the regression coefficients of unrelated components to zero and guarantee the stability of the model. Furthermore, it can prevent false negatives in the

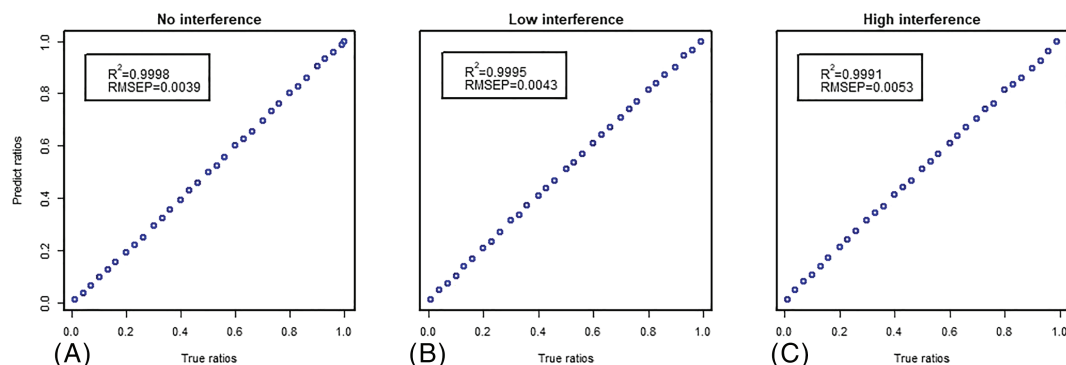


FIGURE 7 The results of ternary quantitative dataset. (A) The plot of the true relative concentration versus predicted relative concentration without interference. (B) The plot of the true relative concentration versus predicted relative concentration with low interference. (C) The plot of the true relative concentration versus predicted relative concentration with high interference (interferences refer to the relative concentration of acetonitrile in the mixture)

TABLE 3 Comparison of NN-EN with PLS, MCR-ALS, and NN-LASSO on the quantitative mixture dataset

		PLS	MCR-ALS	NN-EN	NN-LASSO
No	RMSEP ^a	0.0044	0.0041	0.0039	0.0043
	R^2	0.9854	0.9812	0.9998	0.9841
	PRE (%)	1.45	1.88	0.02	1.59
Medium	RMSEP	0.0061	0.0065	0.0043	0.0056
	R^2	0.9873	0.9853	0.9991	0.9892
	PRE (%)	1.27	1.47	0.09	1.08
High	RMSEP	0.0061	0.0062	0.0053	0.0068
	R^2	0.9870	0.9865	0.9995	0.9840
	PRE (%)	1.30	1.35	0.05	1.60

Abbreviations: NN-EN, non-negative elastic net; NN-LASSO, non-negative lasso; PLS, penalized least squares; PRE, percent relative error; RMSEP, root mean square error of prediction.

^aThe volume ratio of methanol is relative concentration without units.

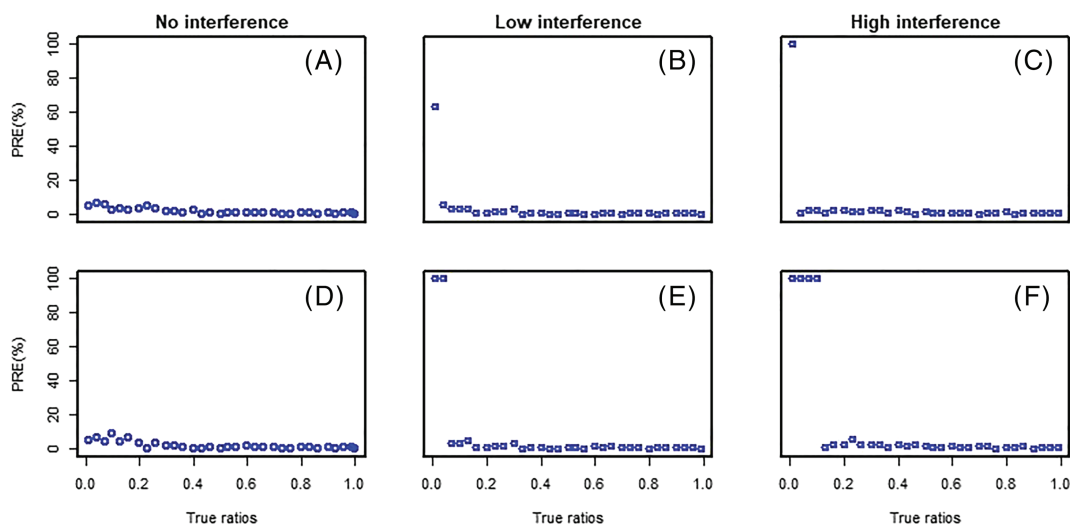


FIGURE 8 Percent relative errors (PREs) obtained by non-negative elastic net (NN-EN) and non-negative lasso (NN-LASSO), respectively. (A–C) The PRE value obtained by NN-EN analysis without interference, low interference, and high interference; (D–F) The PRE value obtained by NN-LASSO analysis without interference, low interference, and high interference (interferences refer to the relative concentration of acetonitrile in the mixture). The PREs of 100% means that the methanol component cannot be identified

identification results due to the quadratic part of the penalty when comparing with NN-LASSO. It can be seen from Table 1 that the NN-LASSO algorithm could not identify the components in the mixture with lower relative concentrations, especially in powder mixtures. But the NN-EN algorithm can accurately identify all the components in mixtures. The PREs of quantitative dataset by NN-EN were shown in Figure 8A–C. The PREs of quantitative dataset by NN-LASSO were shown in Figure 8D–F. The PREs of 100% means that the methanol component cannot be identified. Both NN-EN and NN-LASSO can identify the mixture with methanol relative concentration equals to 0.01 when there is no interference from acetonitrile. When there were interferences, the identification limit of NN-EN (0.01) is much lower than NN-LASSO (0.13). It can be seen from these results that the NN-EN algorithm can accurately identify and estimate the relative concentration of component in the mixture with the existences of interferences when comparing with NN-LASSO.

5 | CONCLUSION

In this paper, the NN-EN method was applied to analyze mixture for Raman spectroscopy. The simulated spectra were analyzed by NN-EN, and it has verified the feasibility of this method. Results on the liquid, powder, tablet dataset, and quantitative dataset of ternary mixtures showed that NN-EN can identify components in mixture and estimate the relative concentrations in real Raman spectra of mixtures, respectively. By comparing with NN-LASSO, one can see that NN-EN can prevent false negatives in the component identification, which is crucial for the mixture analysis. The relative concentrations estimated by NN-EN is comparable with PLS and MCR-ALS. Results show that NN-EN can identify the components in mixture correctly and estimate their relative concentrations with small errors. In the future work, the NN-EN algorithm and state-of-the-art regression techniques will be combined to further increase the capability of the model. It may also be useful for the deconvolution of data-independent acquisition (DIA) in LC-MS.

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